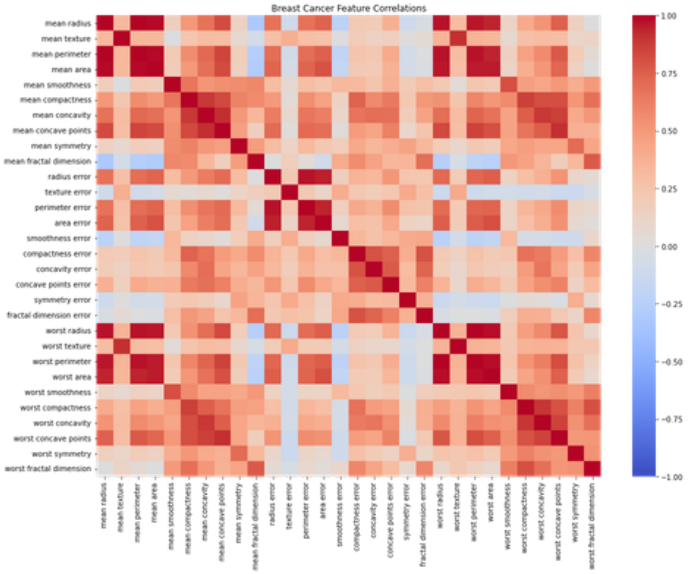
1- The Breast Cancer Wisconsin and Wine datasets were loaded using the load_breast_cancer and load_wine functions in the scikit-learn library. The experiments were conducted using these two datasets.
The original class labels have been preserved. In the Breast Cancer dataset, the label 0 represents malignant cases (212 instances), while the label 1 represents benign cases (357 instances). Malignant cases were designated as the positive class for the malignant case error analysis. In the Wine dataset, the labels 0, 1, and 2 correspond to the classes class0, class1, and class2, containing 59, 71, and 48 examples, respectively.

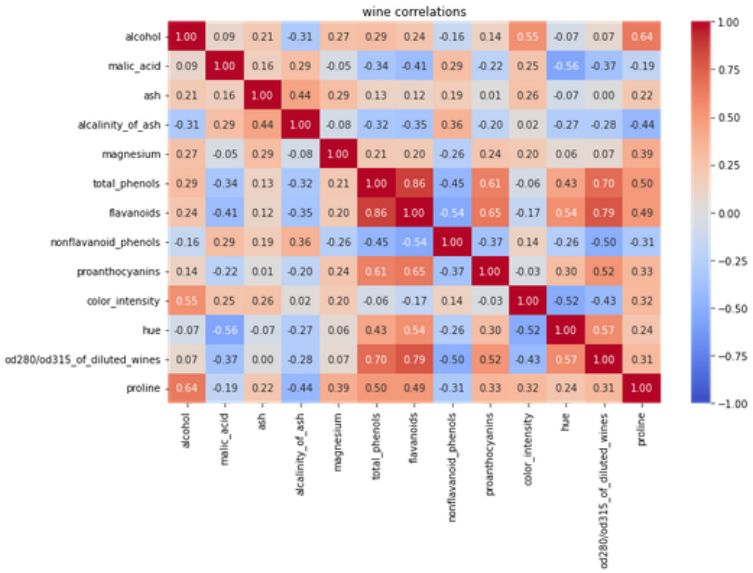
2-Neither dataset contains missing values or duplicate rows.
Potential outliers were identified using the IQR method. Outliers can affect distance-based algorithms such as KNN and K-Means. Decision trees are generally less sensitive to these values.
The Breast Cancer dataset contains 30 numerical features, and the Wine dataset contains 13. Neither dataset contains any categorical features. In both datasets, the data is of the float64 data type.

3-Class distributions show that benign samples are more common than malignant ones. For wines, however, Class 1 appears to be higher than the others. Heat maps of the data are shown below.

Top 3 Correlations:
average radius, average perimeter: 0.997855
worst radius, worst perimeter: 0.993708
average radius, average area: 0.987357
Top 3 Correlations in the Wine Dataset:
total_phenols, flavonoids: 0.864564
Flavonoids in diluted wines: OD280/OD315 0.787194
Total phenols in diluted wines: OD280/OD315 0.699949.
These strong correlations indicate that the data contain similar information.

No features or observations were removed during EDA.





4-The datasets were divided into approximately 70% training data, 10% validation data, and 20% test data. The training data was used for model training, the validation data for selecting appropriate hyperparameters, and the test data for measuring final performance.
The consistency of the results across different splits was examined by using seed values of 42, 52, and 62 in the training and validation splits.
The test set was generated once using the `random_state=42` setting and was not modified.

breast cancer	seed	subset	count	percent	malignant	benign
0	42	train	398	6.995	149	249
1	42	validation	57	1.002	21	36
2	42	test	114	2.004	42	72
3	52	train	398	6.995	149	249
4	52	validation	57	1.002	21	36
5	52	test	114	2.004	42	72
6	62	train	398	6.995	149	249
7	62	validation	57	1.002	21	36
8	62	test	114	2.004	42	72

for: wine	seed	split	samples	percent	class_0	class_1	class_2
0	42	train	124	6.966	41	50	33
1	42	validation	18	1.011	6	7	5
2	42	test	36	2.022	12	14	10
3	52	train	124	6.966	41	50	33
4	52	validation	18	1.011	6	7	5
5	52	test	36	2.022	12	14	10
6	62	train	124	6.966	41	50	33
7	62	validation	18	1.011	6	7	5
8	62	test	36	2.022	12	14	10

5-Since there were no missing values in the Breast Cancer dataset, no value imputation was required.
Due to the wide range of values, StandardScaler was selected during preprocessing to bring the data onto a common scale.

The validation data was transformed using the scaler corresponding to the given seeds; the first five training data points, however, were transformed using only the scaler obtained from seed number 42. To minimize data leakage, we calculated the mean and standard deviation using only the training set.

6-Two versions of the datasets were prepared: one unscaled and one scaled using StandardScaler.

In the wine dataset, both versions used the same 13 features and the same sample indices; only the numerical scales of the features were changed. For each seed, the training data consists of 124 examples. In the breast cancer datasets, seed values of 42, 52, and 62 were used in each dataset. The same features and the same example indices were maintained in both versions.
Thus, the effect of scaling on model performance was compared across the same data splits. A total of 66 models 30 Decision Trees, 30 KNN models, and 6 Gaussiannb models were trained on each dataset, resulting in a total of 132 classification models across the two datasets.

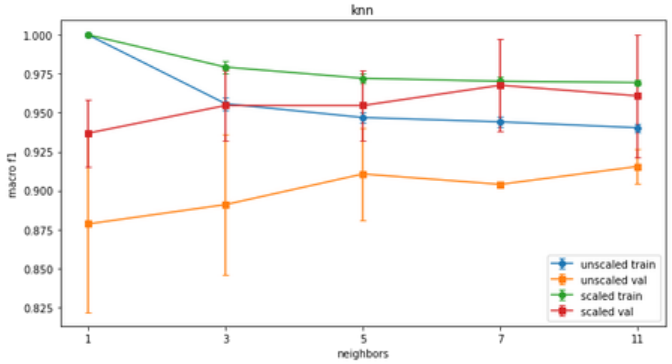
7-For models in the datasets:
Decision Tree = max_depth = 2, 3, 5, 10; number of training runs: 30
KNN n_neighbors = 1, 3, 5, 7, 11; number of training runs: 30
Gaussian Naive Bayes Default settings; number of training runs: 6
Total number of training runs: 66

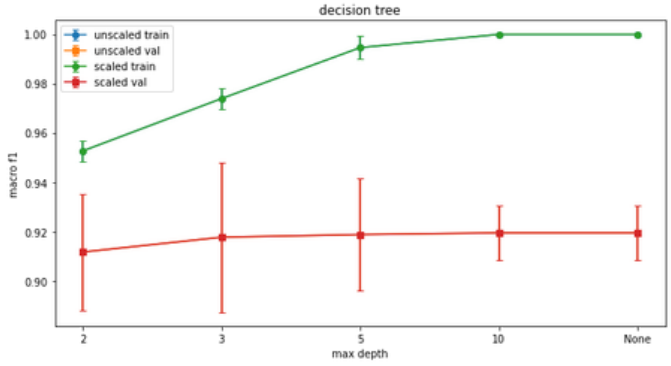
Each model was trained using both scaled and unscaled data and was retrained using the training set with the corresponding seed, repeating the process for settings 42, 52, and 62. The training macro-F1 scores were recorded.
A total of 132 classifiers were trained: 66 for Breast Cancer and 66 for Wine.

8-

	algorithm	representation	parameter	train_accuracy_mean	train_accuracy_std	train_f1_mean	train_f1_std	val_accuracy_mean	val_accuracy_std	val_precision_mean	val_precision_std	val_recall_mean	val_recall_std	val_f1_mean	val_f1_std	val_auc_mean	val_auc_std
0	gnb	scaled	default	0.9355	0.0052	0.9307	0.0058	0.9298	0.0175	0.9239	0.0140	0.9279	0.0289	0.9248	0.0201	0.9757	0.0146
1	gnb	unscaled	default	0.9372	0.0044	0.9323	0.0049	0.9415	0.0203	0.9384	0.0164	0.9372	0.0308	0.9369	0.0226	0.9797	0.0126
2	knn	scaled	1	1.0000	0.0000	1.0000	0.0000	0.9415	0.0203	0.9410	0.0283	0.9339	0.0160	0.9369	0.0213	0.9339	0.0160
3	knn	scaled	11	0.9715	0.0015	0.9694	0.0015	0.9649	0.0351	0.9746	0.0250	0.9524	0.0476	0.9608	0.0395	0.9945	0.0048
4	knn	scaled	3	0.9807	0.0038	0.9793	0.0041	0.9591	0.0203	0.9699	0.0144	0.9444	0.0275	0.9547	0.0228	0.9793	0.0162
5	knn	scaled	5	0.9740	0.0029	0.9721	0.0032	0.9591	0.0203	0.9699	0.0144	0.9444	0.0275	0.9547	0.0228	0.9795	0.0174
6	knn	scaled	7	0.9724	0.0025	0.9702	0.0028	0.9708	0.0268	0.9784	0.0197	0.9603	0.0364	0.9677	0.0297	0.9782	0.0194
7	knn	unscaled	1	1.0000	0.0000	1.0000	0.0000	0.8889	0.0506	0.8836	0.0491	0.8757	0.0630	0.8786	0.0566	0.8757	0.0630
8	knn	unscaled	11	0.9447	0.0025	0.9404	0.0028	0.9240	0.0101	0.9351	0.0145	0.9034	0.0120	0.9155	0.0113	0.9605	0.0111
9	knn	unscaled	3	0.9590	0.0038	0.9558	0.0042	0.9006	0.0405	0.8998	0.0420	0.8849	0.0467	0.8910	0.0449	0.9299	0.0235
10	knn	unscaled	5	0.9506	0.0029	0.9469	0.0032	0.9181	0.0268	0.9174	0.0275	0.9054	0.0310	0.9107	0.0297	0.9480	0.0348
11	knn	unscaled	7	0.9481	0.0029	0.9442	0.0032	0.9123	0.0000	0.9133	0.0066	0.8975	0.0057	0.9040	0.0013	0.9581	0.0291
12	tree	scaled	10	1.0000	0.0000	1.0000	0.0000	0.9240	0.0101	0.9159	0.0112	0.9266	0.0170	0.9195	0.0112	0.9266	0.0170
13	tree	scaled	2	0.9556	0.0038	0.9527	0.0041	0.9181	0.0203	0.9131	0.0161	0.9120	0.0308	0.9117	0.0236	0.9220	0.0404
14	tree	scaled	3	0.9757	0.0038	0.9739	0.0042	0.9240	0.0268	0.9199	0.0231	0.9167	0.0364	0.9177	0.0304	0.9414	0.0209
15	tree	scaled	5	0.9950	0.0044	0.9946	0.0047	0.9240	0.0203	0.9167	0.0182	0.9233	0.0308	0.9188	0.0227	0.9301	0.0250
16	tree	scaled	None	1.0000	0.0000	1.0000	0.0000	0.9240	0.0101	0.9159	0.0112	0.9266	0.0170	0.9195	0.0112	0.9266	0.0170
17	tree	unscaled	10	1.0000	0.0000	1.0000	0.0000	0.9240	0.0101	0.9159	0.0112	0.9266	0.0170	0.9195	0.0112	0.9266	0.0170
18	tree	unscaled	2	0.9556	0.0038	0.9527	0.0041	0.9181	0.0203	0.9131	0.0161	0.9120	0.0308	0.9117	0.0236	0.9220	0.0404
19	tree	unscaled	3	0.9757	0.0038	0.9739	0.0042	0.9240	0.0268	0.9199	0.0231	0.9167	0.0364	0.9177	0.0304	0.9414	0.0209
20	tree	unscaled	5	0.9950	0.0044	0.9946	0.0047	0.9240	0.0203	0.9167	0.0182	0.9233	0.0308	0.9188	0.0227	0.9301	0.0250
21	tree	unscaled	None	1.0000	0.0000	1.0000	0.0000	0.9240	0.0101	0.9159	0.0112	0.9266	0.0170	0.9195	0.0112	0.9266	0.0170

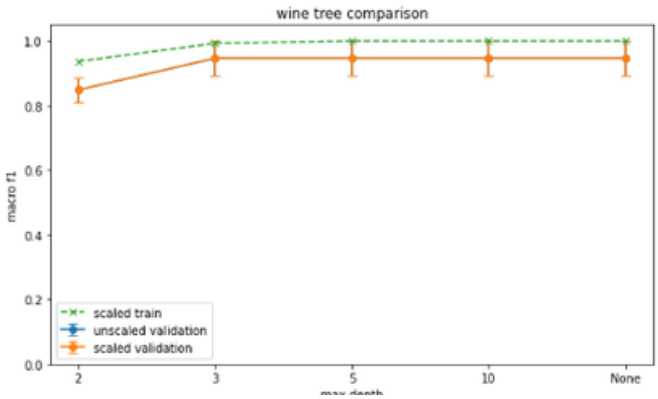
Among the three models used on the breast cancer dataset, KNN the best configuration. Since KNN uses the distances between samples, scaling improved the KNN results. The decision tree results, however, showed very little change with scaling. The best configuration was selected using the mean validation macro-F1 score; the necessary tie-breaking rules were then applied.
best configuration: knn_scaled_7
mean validation F1: 9.677e-01
mean validation accuracy: 0.9708
The model's accuracy was found to be 97.37%, with a macro-F1 score of 0.9713 and an ROC-AUC value of 0.9888. The macro precision of 0.98 and macro recall of 0.9643 indicate high classification performance.

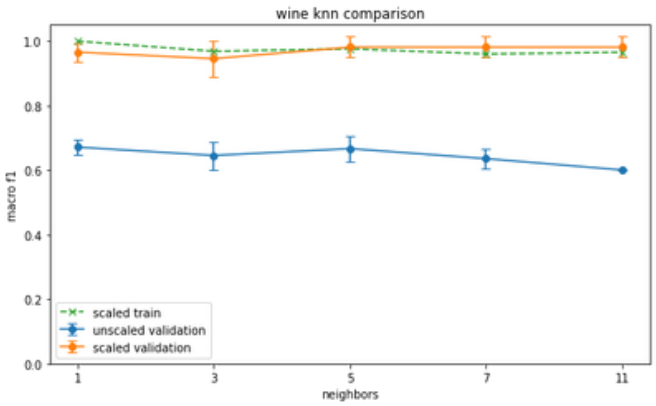




In the Wine dataset, the configuration with the highest average validation macro-F1 score was the KNN model with k = 11, trained on scaled data.
The average macro-F1 score across three seeds was 0.9814, the average accuracy was 0.9815, and the standard deviation of the macro-F1 score was 0.0323.
Therefore, this configuration was selected for the final test evaluation.

	config_id	val_f1_mean	val_f1_std	val_accuracy_mean	val_auc_mean
3	knn_scaled_11	0.981352	0.032299	0.981481	1.000000
5	knn_scaled_5	0.981352	0.032299	0.981481	1.000000
6	knn_scaled_7	0.981352	0.032299	0.981481	1.000000
2	knn_scaled_1	0.965812	0.029608	0.962963	0.974868
0	gnb_scaled_default	0.963844	0.031359	0.962963	1.000000
1	gnb_unscaled_default	0.963844	0.031359	0.962963	1.000000
12	tree_scaled_10	0.946156	0.054515	0.944444	0.955219
14	tree_scaled_3	0.946156	0.054515	0.944444	0.961141
15	tree_scaled_5	0.946156	0.054515	0.944444	0.955219
16	tree_scaled_None	0.946156	0.054515	0.944444	0.955219
17	tree_unscaled_10	0.946156	0.054515	0.944444	0.955219
19	tree_unscaled_3	0.946156	0.054515	0.944444	0.961141
20	tree_unscaled_5	0.946156	0.054515	0.944444	0.955219
21	tree_unscaled_None	0.946156	0.054515	0.944444	0.955219
4	knn_scaled_3	0.945739	0.055809	0.944444	1.000000
13	tree_scaled_2	0.848148	0.036475	0.851852	0.904970
18	tree_unscaled_2	0.848148	0.036475	0.851852	0.904970
7	knn_unscaled_1	0.671386	0.022592	0.703704	0.765361
10	knn_unscaled_5	0.666707	0.039943	0.703704	0.864473
9	knn_unscaled_3	0.645573	0.042877	0.666667	0.833286
11	knn_unscaled_7	0.635589	0.031621	0.666667	0.845858
8	knn_unscaled_11	0.600477	0.000964	0.629630	0.821627

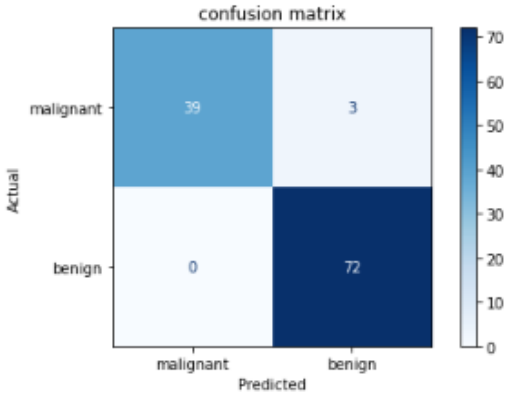




9-The selected breast cancer classifier was evaluated on the validation set using a model trained with a seed value of 42.
The test results were compared with the average validation performance. A confusion matrix was used to examine correct and incorrect predictions

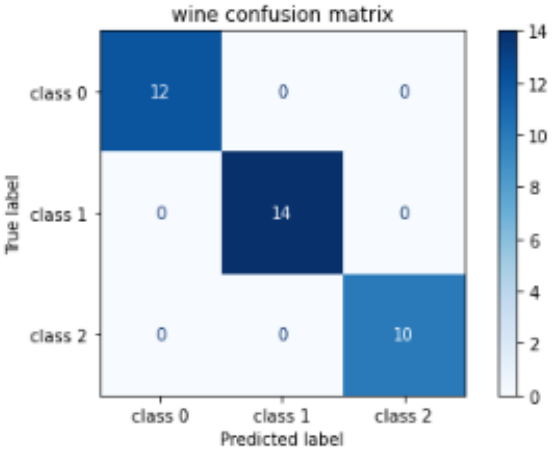
	dataset	configuration	seed	val_f1_mean	val_f1_std	test_accuracy	test_precision	test_recall	test_f1	test_auc
0	Breast Cancer	knn_scaled_7	42	9.677	297	9.737	98	9.643	9.713	9.888

When the malignant class was set as positive in the Breast Cancer dataset, the model correctly identified 39 of the 42 malignant samples and classified 3 as benign, achieving a good success rate.

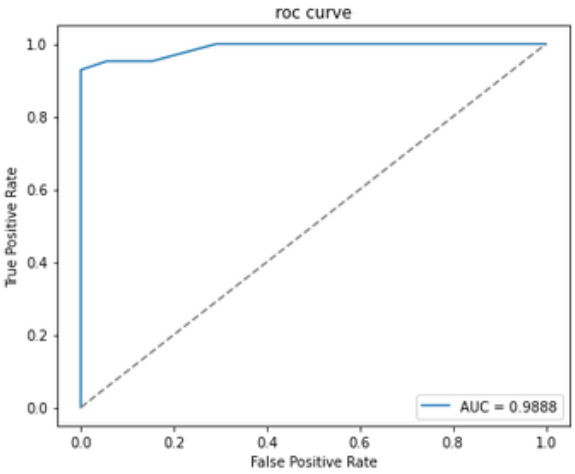


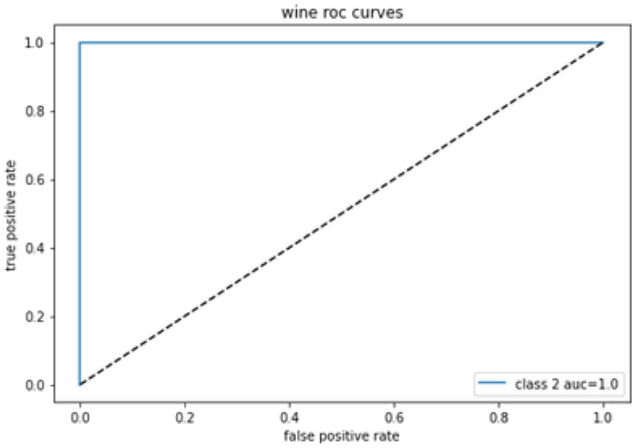
	dataset	configuration	seed	val_f1_mean	val_f1_std	test_accuracy	test_precision	test_recall	test_f1	test_auc
0	Wine	knn_scaled_11	42	981.352	32.299	10	10	10	10	10

All results obtained using the KNN scaled scale with k=11 in the wine set were found to be 1. All samples in the est set were correctly classified. The test macro-F1 value is approximately 0.0186 higher than the validation average.



The ROC curve illustrates the relationship between the true positive rate and the false positive rate at different decision thresholds. A lower threshold may reduce the number of missed malignant cases but may increase the number of false positives.





In the Breast Cancer test set, although the true class of samples with indices 41, 73, and 205 is 0 (malignant), the model predicted these samples as 1. These three errors are classified as “fn” in the confusion matrix table. The model thus missed 3 out of 42 malignant samples.
In the wine set, there are no incorrect predictions.

10-The K-Means method was applied to the datasets using two clusters. Six models were trained using three initial points and two data representations. The models used only training features and did not include class labels.
In this way, we examined the effect of scaling and random initialization on the clustering results.

	dataset	representation	seed	test_ari	test_silhouette
0	Breast Cancer	unscaled	42	5.585	7.120
1	Breast Cancer	scaled	42	7.354	3.251

Models with an initial cluster count of 42 were evaluated on a fixed test set without being retrained.

After scaling the Breast Cancer dataset, the overlap between clusters and actual classes increased. While 28 malignant samples were grouped into the same cluster in the unscaled data, this number rose to 37 in the scaled data. However, 3 benign samples were also included in this cluster.

In the Wine dataset, the classes became more distinctly separated after scaling. All 12 samples in class 0 were grouped in cluster 2, and all 10 samples in class 2 were grouped in cluster 1. Of the 14 samples in class 1, 13 were in cluster 0 and one was in cluster 2.

K-Means did not use class labels during training; the labels were compared only to evaluate the results.

unscaled

cluster	0	1
target		
0	28	14
1	0	72

scaled

cluster	0	1
target		
0	37	5
1	3	69

cluster	0	1	2
target			
0	10	0	2
1	0	9	5
2	0	3	7

scaled

cluster	0	1	2
target			
0	0	0	12
1	13	0	1
2	0	10	0

11-

1-Based on the average validation macro-F1 score, KNN with scaled data performed best on both datasets. While the best setting for the Breast Cancer dataset was 7-neighbor KNN, the best result for the Wine dataset was achieved with 11-neighbor KNN. On the test sets, the Breast Cancer dataset achieved an accuracy of 0.9737, and the Wine dataset achieved an accuracy of 1.0000.
Thus, the same algorithm and representation were selected, but the optimal numbers of neighbors differed.

2-KNN was the algorithm most affected by scaling. For example, in the Breast Cancer dataset, the values increased from 0.9040 to 0.9677. In the Wine dataset, they rose from approximately 0.6005 to 0.9814. Decision Tree results changed very little after scaling.

3-Increasing the tree depth or the number of neighbors did not always yield better results. Deeper trees sometimes produced higher training scores. In the Breast Cancer dataset, scaled KNN with 7 neighbors performed better than KNN with 11 neighbors. The standard deviation of macro-F1 across seeds was 0.0297 in the Breast Cancer data, while in the Wine dataset, this value was 0.0323.

4-The breast cancer classifier made 3 errors: it classified malignant samples as benign. As a result, some malignant cases were missed. In the wine dataset, however, there were no misclassifications.

5-Scaling improved the alignment between the K-Means clusters and the true classes. In the Breast Cancer dataset, the average validation ARI value increased from approximately 0.5043 to 0.5813. In the Wine dataset, this value rose from approximately 0.3613 to 1.0000.

6-One potential improvement for the future could be feature selection. The Breast Cancer dataset contains several features that are similar to one another, and removing unnecessary features could improve performance and reduce redundant information.

For each dataset,
Decision Tree includes 5 depths × 3 seeds × 2 representations = 30 runs;
KNN includes 5 k values × 3 seeds × 2 representations = 30 runs;
GaussianNB includes 1 setting × 3 seeds × 2 representations = 6 runs;
K-Means includes 1 setting × 3 seeds × 2 representations = 6 runs.
This gives 72 runs per dataset and 144 runs across both datasets.

The required figure inventory consists of at least 6 EDA figures, 8 classification figures, and 4 clustering figures, giving a total of at least 18 figure files.

The project walkthrough video is submitted separately to the LMS.

https://github.com/CaganBarutcu7/machine-learning-midterm

File	Row Count
classifier_runs.csv	132
classifier_summary.csv	44
classifier_test.csv	2
kmeans_runs.csv	12
kmeans_summary.csv	4
kmeans_test.csv	4